Metasurface-based Dispersion engineering in Quantum Cascade Lasers for Gas Spectroscopy

Nishant Goel, Gerard Wysocki

Department of Electrical and Computer Engineering, Princeton University

Summary

- We present a metasurface based dispersion engineering method for Quantum Cascade Lasers (QCLs).
- Minimizing dispersion leads to a broad, stable comb operation in QCLs.
- Metasurfaces provide light wavefront control with high degree of freedom.
- Our results show that metasurfaces can reduce dispersion and modify the lasing spectrum.

Effect of Dispersion on QCL Frequency Combs

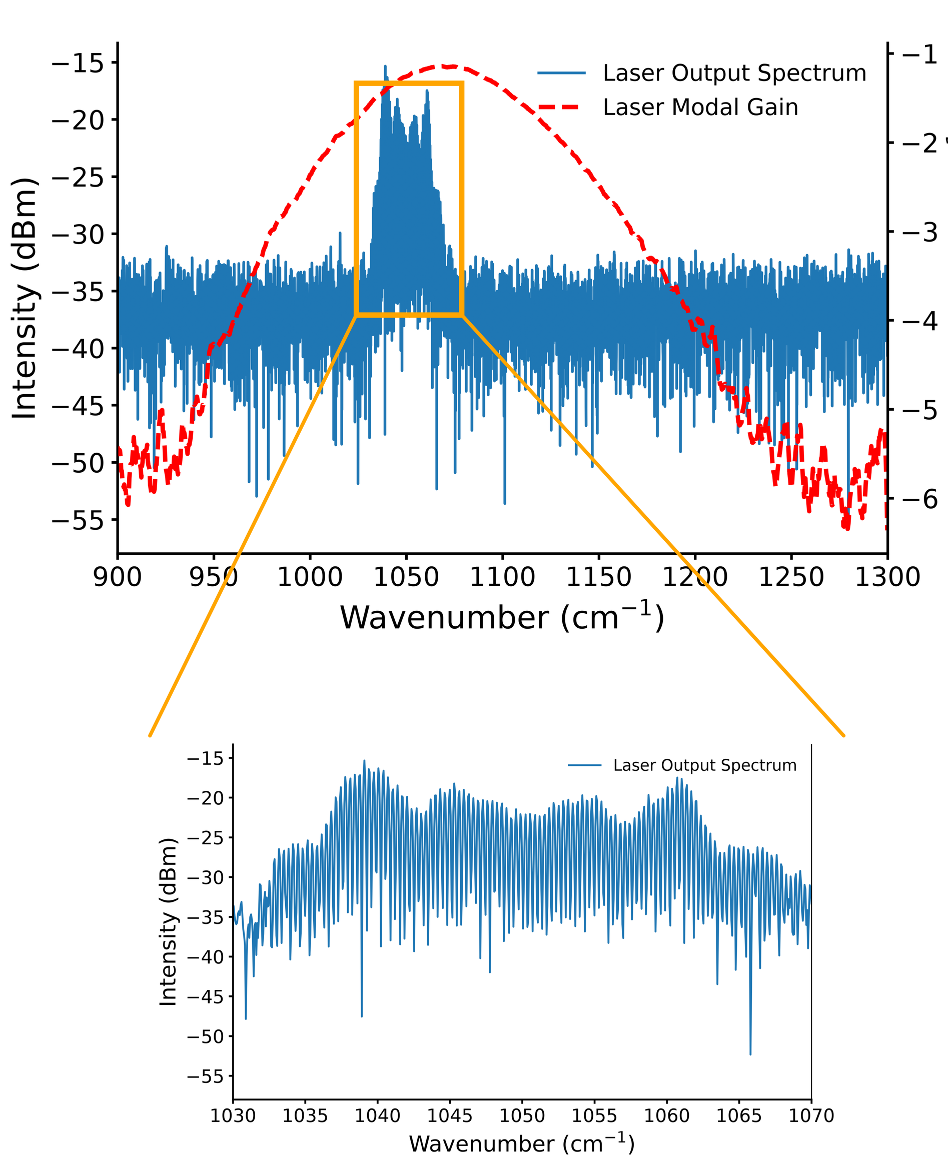

Figure 1: The lasing spectrum of a quantum cascade laser and its modal gain extracted from Hakki-Paoli technique. The lower image shows a magnified view of lasing spectrum, highlighting the distinct comb lines.

The laser's modal gain is calculated using the Hakki-Paoli technique. Dispersion limits the effective utilization of the available gain bandwidth and can lead to:

- Unequal spacing between consecutive comb lines
- Loss of phase locking between comb modes
- Clustering (bunching) of comb lines

Measurement of Dispersion inside QCLs

The Hakki–Paoli technique can be used to extract the intracavity dispersion of a QCL. The laser is operated below threshold to suppress four-wave mixing, which can cause mode pulling and thereby distort the underlying dispersion profile. The dispersion profile contains significant higher-order contributions, making it more challenging to compensate using external dispersion-control mechanisms.

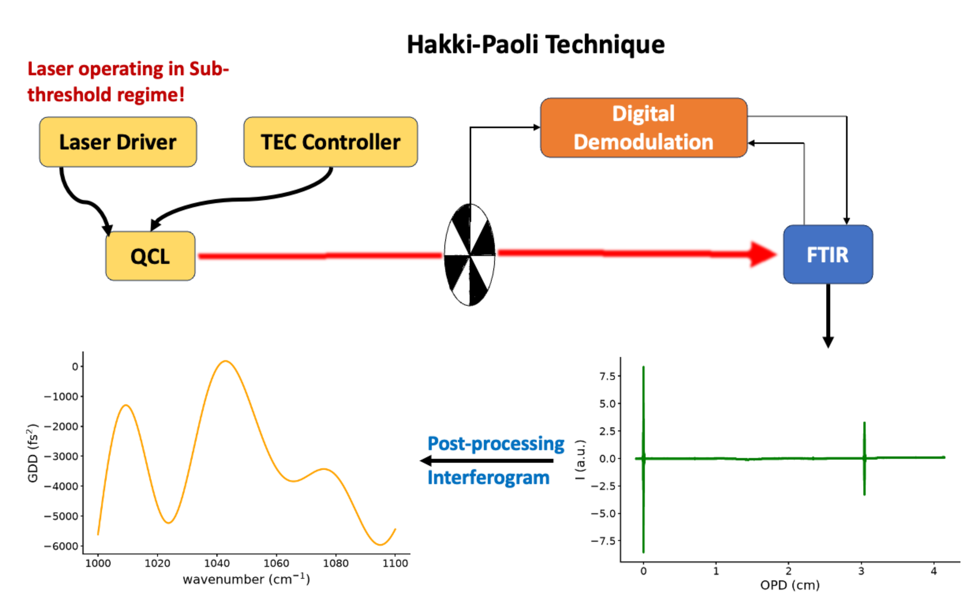

Figure 2: Measurement setup for Hakki-Paoli technique to extract dispersion inside QCL.

Proposed Solution

Dielectric metasurfaces are composed of subwavelength structures that enable precise control of the optical wavefront. For effective dispersion compensation, the metasurface should satisfy the following criteria:

- High mid-IR transmission
- Inverse dispersion of the QCL–external cavity

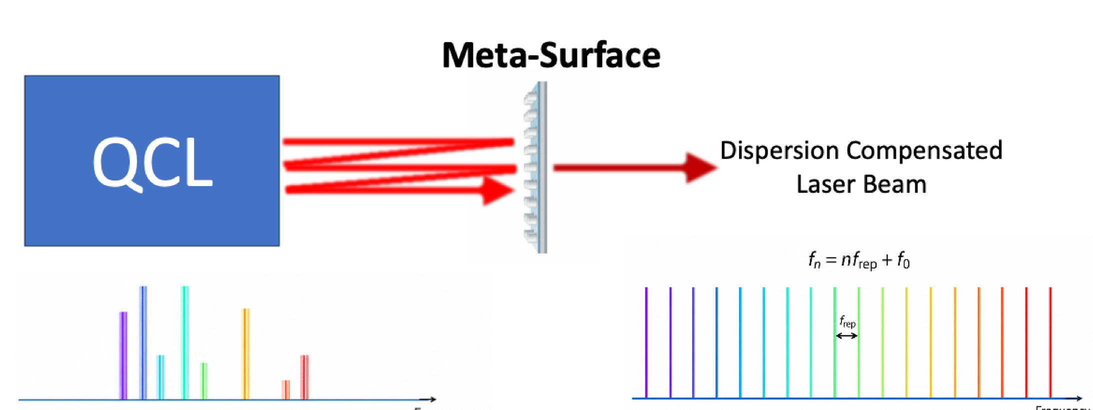

Figure 3: External dispersion compensation using a meta-surface. The frequency spectrum is shown for illustration only, representing an ideal dispersion-free frequency comb.

Fabrication of Metasurface

- a-Si is selected for the meta-atoms due to its low mid-IR absorption.
- An aluminum layer provides a reflective backing on the Si wafer.
- SiO_2 serves as the bonding layer between the aluminum and the 1 μm-thick a-Si layer.
- A chromium hard mask enables sub-3 μm patterning using laser writing.

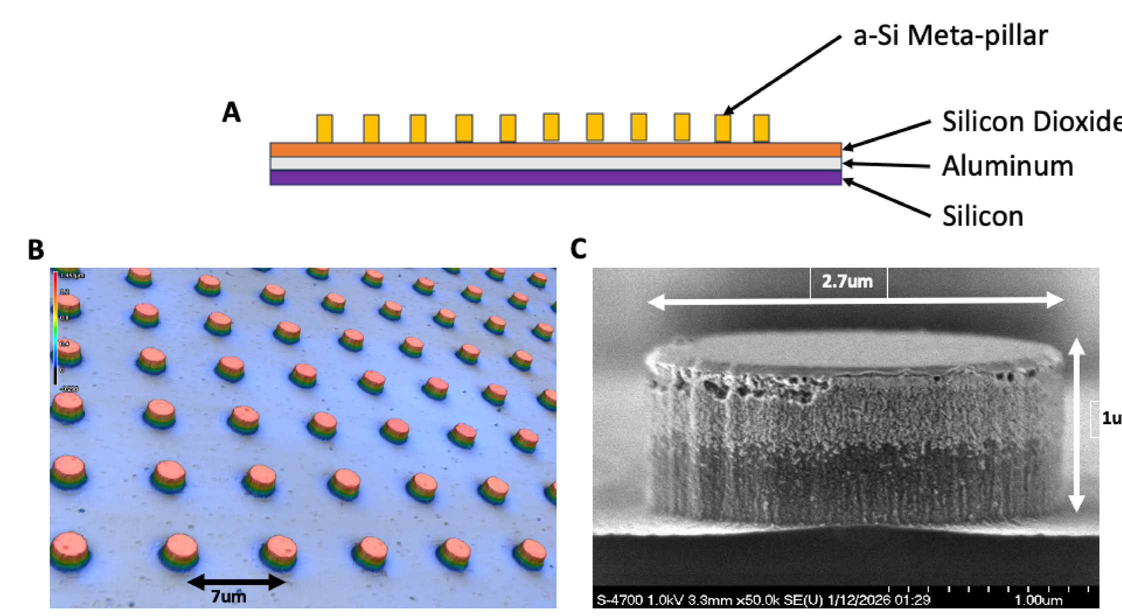

Figure 4: A) Metasurface Layer Stack; B) 3D Image of the Meta-Structure from Confocal Microscope; C) SEM Image of a Meta-Pillar.

Metasurface Characterization

Although the metasurface dispersion cannot be measured directly, its FTIR reflection spectrum can be compared with FDTD simulations.

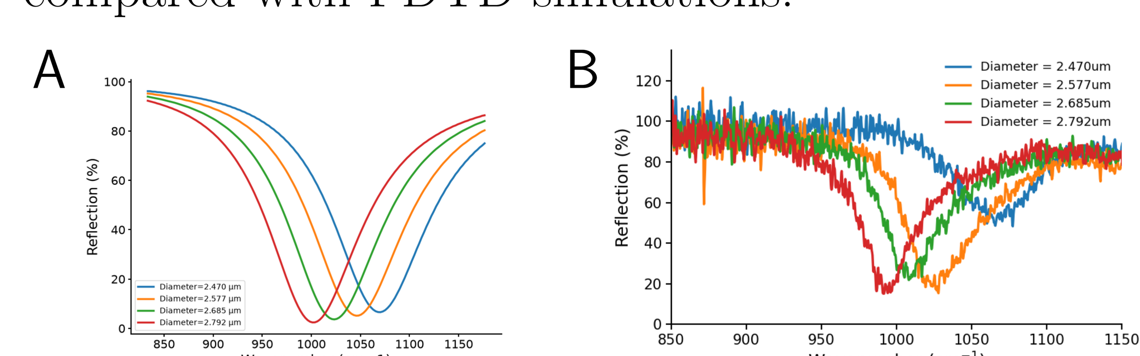

Figure 5: Reflection characterization of metasurfaces with varying diameters: (A) FDTD simulations and (B) experimental FTIR measurements.

As shown in Fig. 5, the measurements closely match the simulations, with small deviations likely due to uncertainties in deposited layer thicknesses, differences in the actual material refractive indices, and the limited precision of laser writing at these feature sizes.

Phase-locked laser emission with Metasurface

A shoulder-free electrical intermode beatnote indicates phase locking in the QCL. The coupled QCL–metasurface system exhibits phase-locked operation, although with a slight reduction in lasing bandwidth. This reduction is primarily attributed to the metasurface dispersion being optimized over a relatively narrow spectral range.

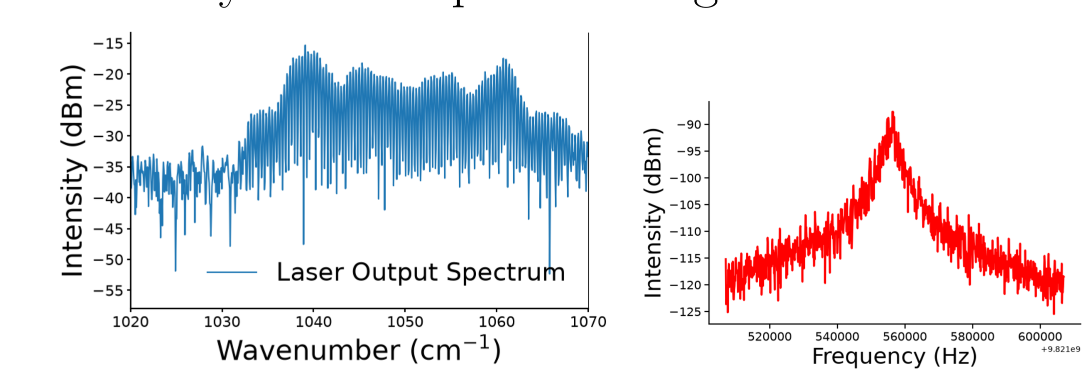

Figure 6: Free running laser spectrum (left) and its corresponding electrical intermode beatnote (right).

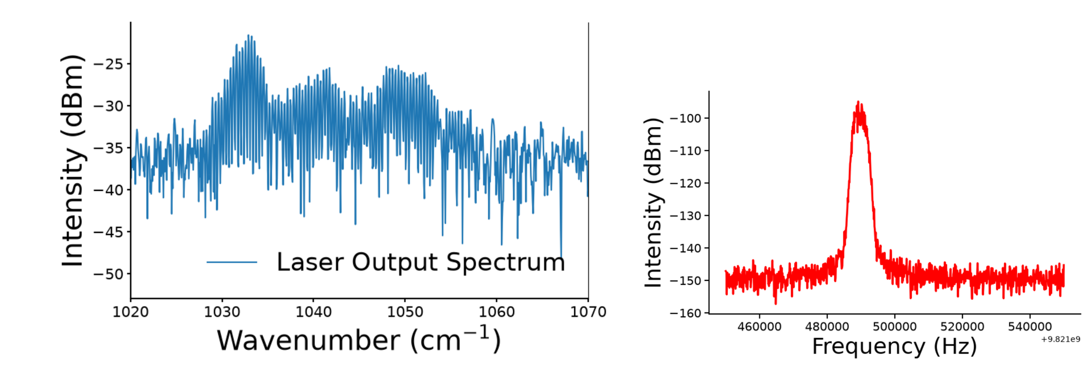

Figure 7: Laser spectrum from the coupled QCL–metasurface system (D = 2.3 μm) (left) and the corresponding electrical intermode beatnote (right).

References & Acknowledgement

[1] Hofstetter, D., & Faist, J. (1999) IEEE PTL.
[2] Hugi, A. et al. (2012) Nature.
[2] Khorasaninejad, M., et al. (2016) Science.
[3] Opačak, N., & Schwarz, B. (2019) PRL.

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